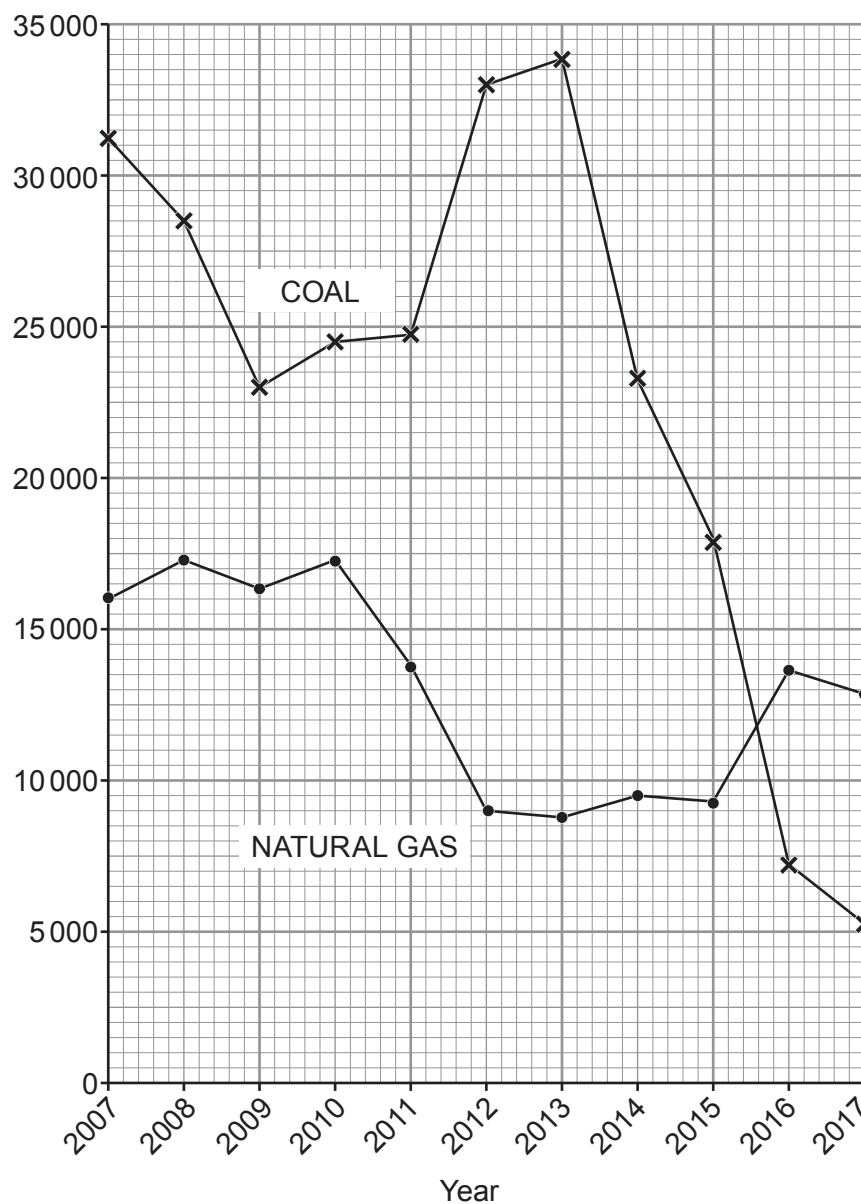


Answer **all** questions.

1. The UK government wants to reduce the CO₂ emissions from power stations.

The graphs below show the CO₂ emissions from coal and natural gas power stations between the years 2007 and 2017.

CO₂ emissions (kilotonnes)



- (a) Look at the data for **2012**. Calculate the difference in the emissions of CO₂ from coal and natural gas. [2]

difference = kilotonnes of CO₂

- (b) Between the years 2014 and 2015, the emission of CO₂ from coal fell by 5500 kilotonnes.

State between which other years the emission of CO₂ from coal fell **at the same rate**. [1]

Years and

- (c) State **two** benefits of reducing CO₂ in the atmosphere. [2]

1.

2.

- (d) (i) Nuclear power stations provide up to 20% of the present UK demand for electricity.

Gas provides up to 50%.

One student, Seren, says that a graph for the CO₂ emissions from nuclear power stations would be the same shape as for gas but always lower.

Explain whether you agree with Seren. [2]

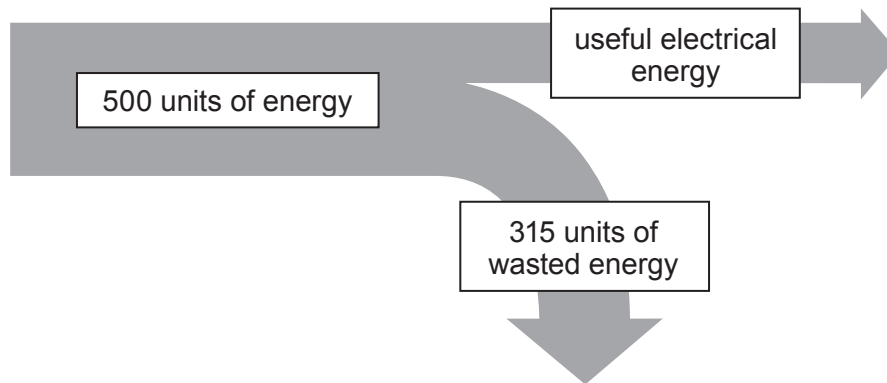
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- (ii) The Sankey diagram shows the energy input and output for a power station.



Seren looks at the diagram and calculates that the power station is 63% efficient.

Explain whether you agree with Seren.

[2]

Space for calculation.

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